

(1930) $f_1, f_2, \dots$ independent, same distribution

Assume $\mathbb{E}(|f_1|) < \infty$ ^(*), then: $\frac{1}{N} \sum_{n=1}^N f_n \xrightarrow[N \uparrow \infty]{a.e.} \mathbb{E}(f_1)$

Strong Law of Large Numbers

Conversely, if $\frac{1}{N} \sum_{n=1}^N f_n \xrightarrow[N \uparrow \infty]{a.e.} 0$, then $\mathbb{E}(|f_1|) < \infty$ and $\mathbb{E}(f_1) = 0$.

(1925, 28) Kolmogorov - ~~Sk~~ Lincin

Assume $\lim_{M \rightarrow \infty} (M \cdot P(|f_1| > M)) = 0$ (weaker than ^(*))

Then: $\frac{1}{N} \sum_{n=1}^N f_n - \underbrace{\mathbb{E}[f_1 \mathbb{1}_{\{|f_1| \leq n\}}]}_{D_N} \xrightarrow[N \rightarrow \infty]{P} 0$

Weak Law of Large Numbers

Reference: Feller, Vol II., pp. 265-266

KOMLÓS (1967) $f_1, f_2, \dots$ real-valued measurable

Assume $\sup_{n \in \mathbb{N}} \mathbb{E}^P(|f_n|) < \infty$ ^(*), then there exists a subsequence

$f_{k_1}, f_{k_2}, \dots$ and $f_*: \Omega \rightarrow \mathbb{R}$, $\mathbb{E}(|f_*|) < \infty$, so that:

$$\frac{1}{N} \sum_{n=1}^N f_{k_n} \xrightarrow[N \rightarrow \infty]{\text{a.e.}} f_*$$

A. Rényi: " f_* behaves like an expectation"

also along any further subsequences.

Walter Schachermeyer, István Berkes: Is (*) necessary?

Answer: NO!

↳ Depends on requirements about f_* .

Suppose for $f_1, f_2, \dots$ measurable, there exists $f_*: \Omega \rightarrow \mathbb{R}$ measurable and $f_{k_1}, f_{k_2}, \dots$ s.t. Then, there exist

$A_1 \subseteq A_2 \subseteq \dots \subseteq A_j \subseteq \dots$ with $\lim_{j \rightarrow \infty} P(A_j) = 1$ and

$\sup_{n \in \mathbb{N}} \mathbb{E}^P[|f_{k_n}| \cdot \mathbb{1}_{A_j}] < \infty, \forall j \in \mathbb{N}$. And v.v.:

$\sup_{n \in \mathbb{N}} \mathbb{E}^Q(|f_{k_n}|) < \infty$, for some $Q \sim P$.

If requires $\mathbb{E}[|f_*|] < \infty$, the necessary and sufficient

condition: $\exists A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq \dots$ $\lim_{n \rightarrow \infty} P(A_n) = 1$
s.t. $\sup_{n \in \mathbb{N}} \mathbb{E}^P(|f_{k_n}| \mathbb{1}_{A_n}) < \infty$

Regarding weak law: $f_1, f_2, \dots$ measurable

Assume $M \cdot \sup_{n \in \mathbb{N}} P(|f_n| > M) \xrightarrow{M \rightarrow \infty} 0$. Then, $\exists f_{k_1}, f_{k_2}, \dots$

Random correctors $D_1, D_2, \dots$, $P(|D_N| \leq N, \forall N \in \mathbb{N}) = 1$

and $\frac{1}{N} \sum_1^N f_{k_n} - D_N \xrightarrow[N \rightarrow \infty]{P} 0$.

